

Philosophy :-

- "The region of liberating doubt"
- "philosophy is a perspective on life."
- "that which questions assumption"

Philosophy etymologically → love of wisdom

Humans are ~~curious~~ curious to find the Truth of something
 Philosophy is the quest for Truth, asks for reasons.
 History of phil → history of questioning, of not taking things for granted

Science is descriptive, Philosophy is normative (?)

Philosophy of science questions the methods & biases of science itself.

Antidote to Dunning - Kruger Effect.

"The unexamined life is not worth living"

Description ← "What is the case?"

Normative ← "What ought to be the case?"

Philosophy tries to examination of argumentation of others,
 it emerges in disagreement

Logic?

Philosophy uses logic, reasoning, inference, premises, etc
 while science uses observations, calculations, experiment etc.

Inferences have (hidden) premises.

We can't do without logic

Emotions also sway logic.

Informal logic $\Rightarrow$ everyday reasoning.

Fallacy - Reasoning that looks sound but upon further examination found to not be logical.

Law of identity: Logical arguments / debates only if the topic of the discussion is defined the same by both all parties involved. One can't define as X while the other defines as Y.

Terms must be set to have a coherent resolution to any disagreement.

Hard to prove absot. universal negation, hence the burden of proof lies with the believer.

Argument = Conclusion + Premise, good when the truth of the premise leads to the truth of the conclusion through inference.

Deductive & Inductive inferences

Sentence vs Proposition

Premises $\Rightarrow$ Proposition

Propositions always assert something. Can only be boolean.

Sentences need not assert. Can not have a truth value.

Truth & Validity

$\downarrow$ $\hookrightarrow$ for arguments

for propositions

Argument can be valid with false proposition

Soundness apt applies to deductive arguments if the premises are true and the deduction is valid.

Soundness is not binary, if we are not sure of the truthfulness of the premises.

Deductive Arg:-

Conclusion preserves the truth of the premises.

② Laws of thought:-

(i) Law of non contradiction cannot be $A \wedge \neg A$

→ (ii) Law of identity $A = A$

(iii) Law of excluded middle A can only be $A \vee \neg A$

Quantum Physics breaks (i) (to some extent) & (iii)

Aristotelean proof for (i) used reductio ad absurdum.

Non contradiction & paradoxes:

Physically impossible but logically possible.

But what about that which is physically possible but logically impossible
Quantum physics

Deduction is non-ampliative → gives ~~new~~ ^{no new} information, all info exists a priori in the premises

Induction

→ Only probabilistic truths.

No new premise in deduction arguments changes the validity.

Science is inductive.

gng ts not tuff

never

classmate

Date _____

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New premises might change our conclusions.

Not Truth Preserving

Validity does not apply to Inductive Arguments

Inductive Arguments can only be weak or strong

New data/evidence can always appear.

Degree vs Certainty

Inductive vs Deductive

Strong Inductive vs Weak Induction

↓

High likelihood
of conclusion given
premises

↓

low likelihood of
conclusion given premises

DO NOT READ BEYOND WHAT IS GIVEN. ALL PREMISES
MUST BE STATED, ONLY THAT WHICH IS STATED IS A VALID
PREMISE

Antecedent is sufficient for the consequent, the consequent
is necessary for the antecedent

Induction is circular, it depends on itself for its existence
Depends on the Principle of Uniformity of Nature

Fallacies (Informal)

Sophistry vs Fallacy

↓

Delude the masses by
using fallacies ← Rhetoric

(i) Fallacy of Relevance:

Premises must be relevant to the conclusion

Fallacies deal with the semantics / meaning of the argument

We should be vigilant about rhetoric

(a) Appeal to Populace :-

Majority view is not always right

Common Moral Truth vs Universal Moral Truth



statistically most cultures follow this



regardless of preconditions this is true

(b) Appeal to Emotion

(c) Red Herring

Whataboutism

Distraction & Diversion based rhetorical tactics

(d) Strawman :-

↑ straw man

Convert the opponent's argument into its weakest version and then attack it.

Convert all premises into their weakest version.

Red herring does not care about the premises at all

Principle of Charity :- Core belief in philosophy : Take the strongest possible version of an argument and try to refute it.

⑥ Ad Hominem

Attack the person not the argument. Can be true or -ve.
 Just • Basically one ought not make judgments of people's claims based on
 Abuse or Circumstantial. their association

↓

Character Assassination

Tu Quoque
~~Quoque~~

Guilt by Association

↳ "You did it too!"

Poisoning the well.

↳ Attacking Motivations

↳ "This person is anti-AI because
they're from the humanities"

⑦ Missing The Point:-

Catch-all phrase.

The conclusion is at best vaguely related to the conclusion supported
 by the premises.

Conclusions here are usually derived from one single premise

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